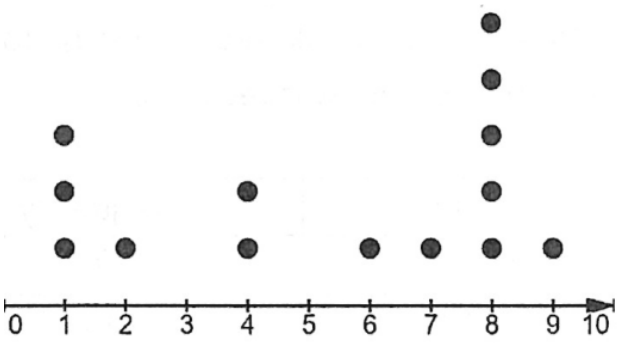
Representing Data

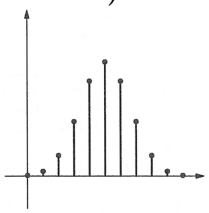
Questions 1–16

Question 1. Find the mean from the dotplot:

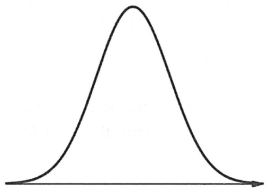


Question 2. Use the terms: uniform, unimodal, bimodal, continuous, discrete, symmetric, positively skewed, and/or negatively skewed to describe:

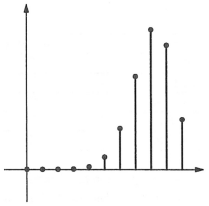
a)



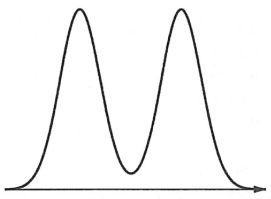
b)



c)

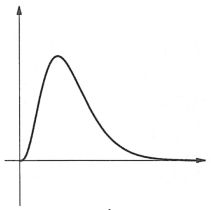


d)

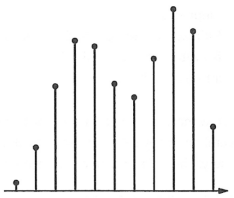


Question 2 (continued). Use the terms: uniform, unimodal, bimodal, continuous, discrete, symmetric, positively skewed, and/or negatively skewed to describe:

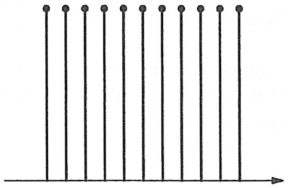
e)



f)



g)



Question 3. The stem-and-leaf diagram shows the weights of a sample of 29 male moose measured to the nearest kg.

38	1	2	5	6	6			
39	3	3	4	7	8	9	9	9
40	0	0	2	2	3	8		
41	0	1	2	3	9			
42	2	2	2					
43	0	1						

Key: 38|1 represents 381 kg

a) Describe the shape of the distribution.
b) Find the median.
c) Find the interquartile range.

Question 4. The stem-and-leaf diagram shows the heights of a sample of 13 red kangaroos measured to the nearest cm.

7	2 3 3
8	1 2 3 6 7 8
9	5 5
10	0
11	
12	9

Key: 7|4 represents 74 cm

a) Find the median and interquartile range of the red kangaroo heights.
b) Calculate the mean and the standard deviation of the red kangaroo heights.
c) Explain why the median would be a better measure of central tendency than the mean in this case.

Question 5. The stem-and-leaf diagram shows the wingspans of a sample of 23 Arctic terns measured to the nearest cm.

31	0
32	8 9
33	0 1 2 4
34	0 0 4 4 8
35	3 3 7 8 8 9
36	0 0 0 1 1

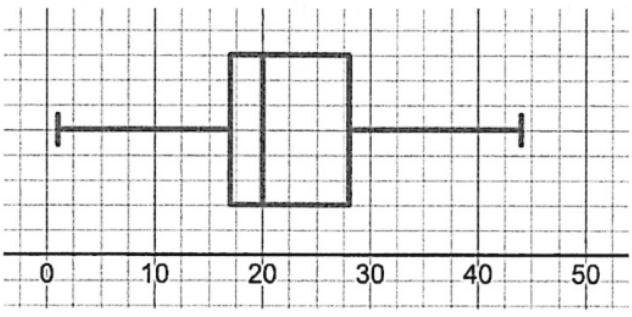
Key: 31|0 represents 31.0 cm

a) Describe the shape of the distribution.
b) Alan looks at the data in the stem-and-leaf diagram and determines that 36.0 cm is the average wingspan of an Arctic tern. Explain how Alan can justify this result.
c) Brian looks at the data in the stem-and-leaf diagram and determines that 34.8 cm is the average wingspan of an Arctic tern. Explain how Brian can justify this result.
d) Draw a boxplot to represent the data.

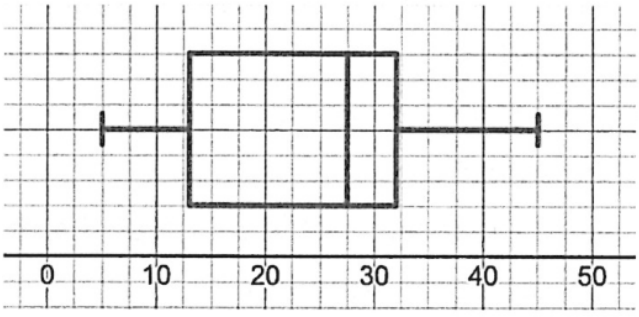
Question 6. Draw a boxplot with negative skew.

Question 7. For the following boxplots, identify whether they are roughly symmetric, positively skewed or negatively skewed:

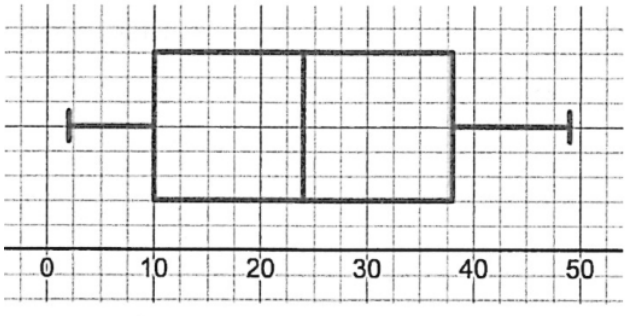
a)



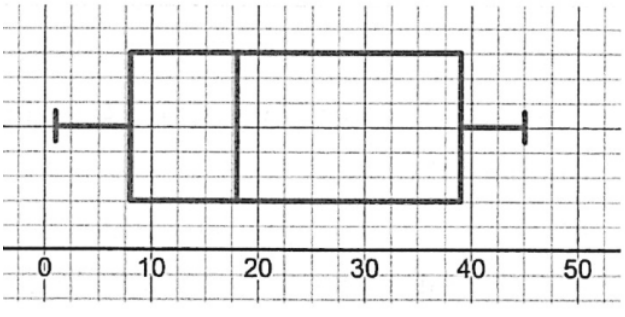
b)



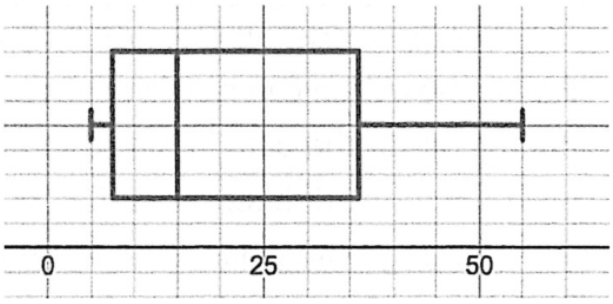
c)



d)

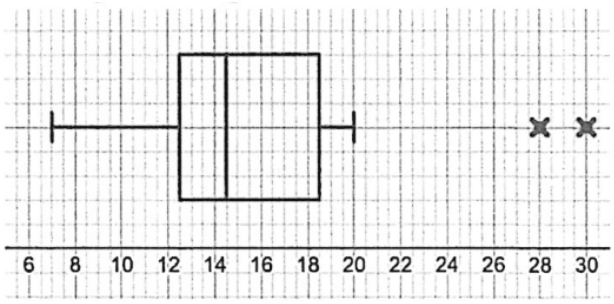


Question 8. The boxplot below represents the commute times, in minutes, of employees at a large company.



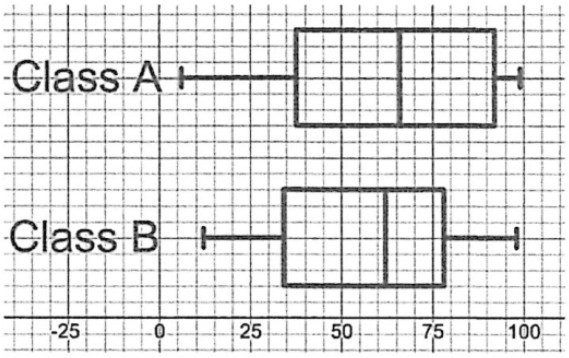
a) State the name of the measure of central tendency identified by the box plot, and its value.
b) Determine whether the box plot is symmetric, negatively skewed or positively skewed.

Question 9. The boxplot below represents the number of dogs that attended puppy training each Thursday evening for 20 weeks.



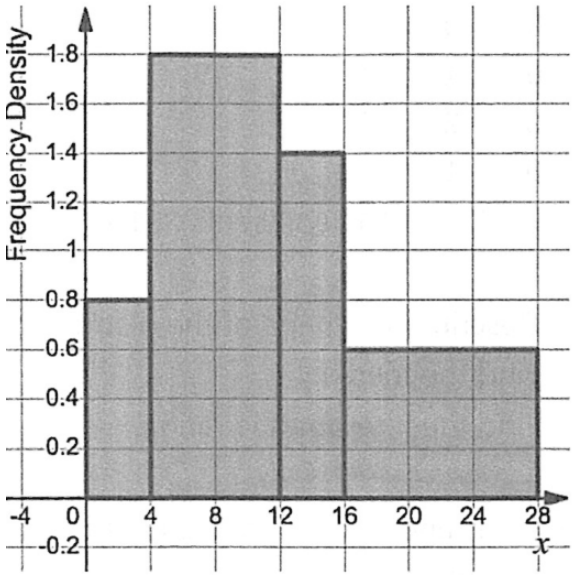
a) Find the median number of dogs that attended.
b) Find the range of the number of dogs that attended.
c) Find the interquartile range of the number of dogs that attended.

Question 10. Two classes sat a test where the maximum score was 100. The results are displayed in the two boxplots below.



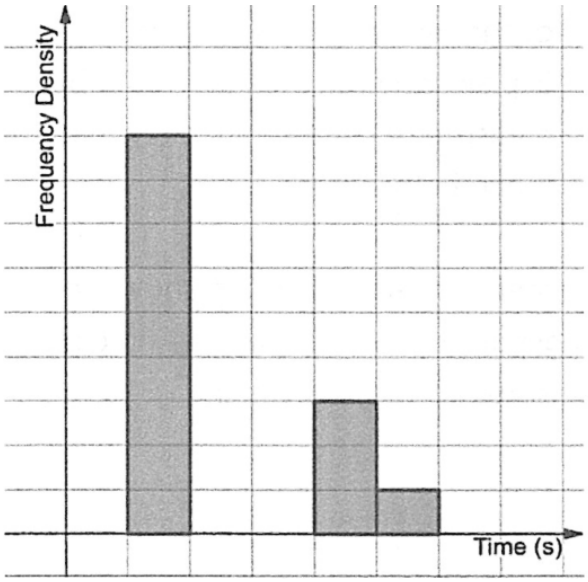
a) Estimate the median of Class A and the median of Class B.
b) Estimate the interquartile range of Class A and the interquartile range of Class B.
c) Use your answers to compare the results of the two classes.

Question 11. The histogram below shows the times, x minutes, that employees at a company take to walk to work. Estimate the proportion of employees that take between 9 minutes and 25 minutes to walk to work.



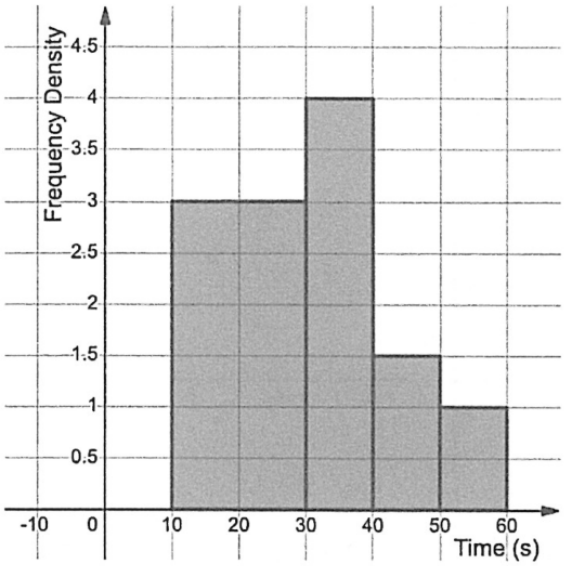
Question 12. Aminah is investigating the length of time, t minutes, meetings take at her place of work. She is partway through drawing a histogram from the following table of data:

Meeting Length (t)	Frequency
$0 \leq t < 30$	30
$30 \leq t < 60$	45
$60 \leq t < 120$	20
$120 \leq t < 150$	15
$150 \leq t < 180$	



a) Copy and complete the table and histogram.
b) Estimate the number of meetings that last between 40 and 99 minutes.

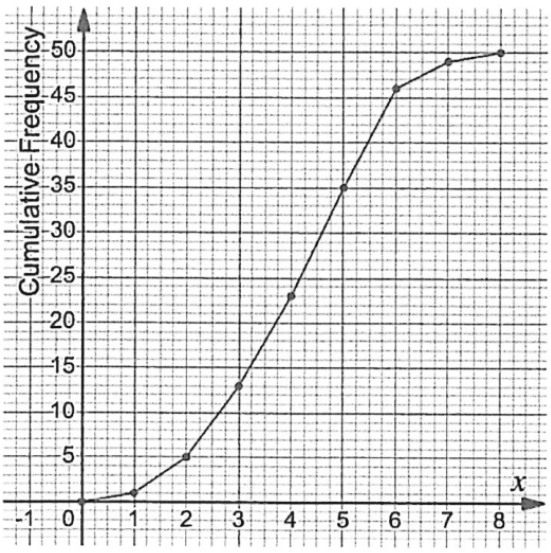
Question 13. The histogram below shows the times, in seconds, for a sample of adults to complete a puzzle.



a) Which class interval contains the median?
b) Estimate the mean.
c) Estimate the standard deviation to 3sf.
d) An adult is selected at random. Find the probability that they took between 28 and 32 seconds to complete the puzzle.

Question 14. Umar took a random sample of 24 apples and they weighed between 160 and 180 grams. He drew a histogram to represent the data and the bar representing 160 to 180 grams has a width of 2 cm and a height of 6 cm. There are 16 apples that weigh between 180 and 210 grams. Determine the width and height of the bar on the histogram representing 180 to 210 grams.

Question 15. Use the following cumulative frequency graph to estimate the median and interquartile range.

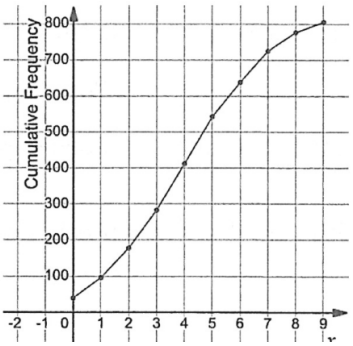


Question 16. Amber sells ice-creams. One year, she counts the number of customers she sells ice creams to over ten weeks of Sundays in the summer. Her sales are shown in the table below:

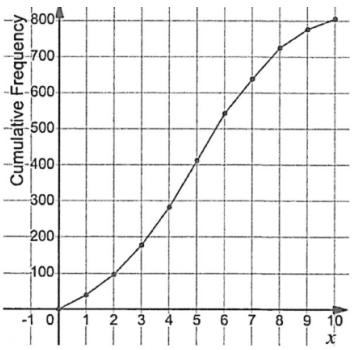
Week	0 –	1 –	2 –	3 –	4 –
Sales	40	56	82	105	129
Week	5 –	6 –	7 –	8 –	9 –
Sales	131	95	87	51	29

a) Amber draws two different cumulative frequency graphs. These are shown below. Which of the two should she use to represent her data? Explain your reasoning.

Graph A:



Question 16 (continued). Graph B:



b) Amber shows the correct graph to Terry who says that it shows that week on week her sales are increasing. With reference to the graph, explain whether Terry is correct.

c) Terry goes on to say that Amber can use the correct graph to estimate the exact time during a week that Amber had sold half of the ice creams. Explain why this would be inappropriate.